

ViterbiNet Revisited: A Matched-Information Classical Baseline for Learned Trellis Detection

Gil Zukerman

Abstract—ViterbiNet replaces the channel-dependent branch metric of the Viterbi algorithm with a small neural network trained from pilots and adapted online on decoded words. Its reported advantage is measured against a Viterbi detector given perfect or deliberately corrupted channel state information (CSI). We add the missing baseline: a Viterbi detector that receives exactly the information ViterbiNet receives (a pilot word every 25 words and the same code-gated decisions) and estimates its four taps by least squares. On a coded BPSK intersymbol-interference channel with COST 2100 taps, this LS-Viterbi receiver has the lowest point-estimate SER of all no-CSI receivers from 0 to 14 dB. On shared Monte Carlo draws it is significantly better than the best ViterbiNet configuration at 6, 7, and 9–12 dB and never significantly worse; at 13–14 dB the comparison is not statistically resolved. It stays within 3–30% of the perfect-CSI SER, consistent with a predicted 0.13 dB least-squares estimation loss. Replacing the reference implementation’s oracle adaptation gate with an implementable, ground-truth-free one changes no SER by more than its confidence interval. We further show that ViterbiNet’s default of 200 online gradient steps per word overfits (5 steps gives a lower SER at every SNR and is 40 times cheaper), and that a 32-parameter affine metric, the exact form of the Gaussian log-likelihood, matches the 7,002-parameter network. Only LS-Viterbi and ViterbiNet with 5 steps meet a real-time budget on one CPU thread.

Index Terms—Deep learning, symbol detection, Viterbi algorithm, channel estimation, intersymbol interference, model-based deep learning.

I. INTRODUCTION

MODEL-BASED deep learning keeps the structure of a classical algorithm and learns only the parts that depend on an unknown model [1]. ViterbiNet [2] is a representative example for sequence detection over channels with memory: the Viterbi trellis [3], [4] is kept, and a small fully connected network maps each received sample to the likelihoods of the channel states. The network is trained on pilots and adapted online on decoded words that a Reed–Solomon (RS) code accepts, which lets the receiver follow a time-varying channel without new pilots. Later work added meta-learning to speed up that adaptation [5], [6]. More broadly, learned detectors have been proposed for the physical layer [7], OFDM [8], MIMO [9], and channels without tractable models [10].

ViterbiNet’s performance is reported relative to the Viterbi algorithm with full CSI, which it approaches, and with CSI uncertainty, which it outperforms [2]. Neither baseline receives the information ViterbiNet receives. The first is given the true channel; the second is given a corrupted one and no means of correcting it. A classical receiver that estimates the channel

from the same pilots and the same accepted decisions is the matched comparison. Such adaptive maximum-likelihood sequence estimators are decades old [11], [12], but they are absent from the evaluations of learned trellis detectors that we are aware of.

This letter makes three contributions:

- A matched-information baseline, LS-Viterbi, which is never worse than ViterbiNet on paired draws, is significantly better at most SNRs from 6 to 12 dB, and tracks the perfect-CSI bound within a loss we predict (Section III-A).
- Evidence that ViterbiNet’s online adaptation overfits: 5 gradient steps per accepted word outperform the default 200 (Section III-B).
- A 32-parameter affine branch metric that matches the 7,002-parameter network, and a cost comparison against a real-time budget (Section III-C).

II. SYSTEM MODEL AND RECEIVERS

A. Channel and frame

Information words of 120 bits are RS-encoded over $\text{GF}(2^8)$ with two parity symbols [13], BPSK-modulated ($s_t \in \{\pm 1\}$), and sent over a real intersymbol-interference channel with memory $L = 4$:

$$y_t = \sum_{k=0}^{L-1} h_k^{(w)} s_{t+L-1-k} + n_t, \quad n_t \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

where $\text{SNR} = 1/\sigma^2$. The taps $h_k^{(w)}$ of word w are the magnitudes of a fixed COST 2100 channel trace [14] (300 blocks; within each block of 50 words the taps decay smoothly, and between blocks they jump, so every frame contains two abrupt changes); $L = 4$ matches the intersymbol-interference setting of [2]. A frame has 125 words; every 25th word is a pilot, so each frame carries 5 pilots and 120 data words. The trellis has $2^L = 16$ states $x_t = (s_t, \dots, s_{t+L-1})$.

B. Receivers

All receivers run the same 16-state Viterbi recursion and differ only in the branch metric $\ell_t(x)$ assigned to state x at time t .

Viterbi, perfect CSI: $\ell_t(x) = -(y_t - \mathbf{h}^T \mathbf{s}(x))^2/2$ with the true taps. This is the lower bound.

Viterbi, noisy CSI: the same metric with taps $1, \dots, L - 1$ perturbed by Gaussian noise whose standard deviation is a fraction ρ of the RMS tap magnitude (the CSI-uncertainty baseline of [2]).

LS-Viterbi (matched baseline, no CSI): the same metric with $\hat{\mathbf{h}} = \arg \min_{\mathbf{h}} \sum_t (y_t - \mathbf{h}^T \mathbf{s}(x_t))^2$, recomputed on every pilot word and on every data word that passes the adaptation gate, with x_t taken from that word's labels. The gate and the labels are those of the learned receivers, applied identically to every adaptive receiver.

ViterbiNet: a network $1 \rightarrow 100 \rightarrow 58 \rightarrow 16$ (sigmoid, then ReLU; 7,002 parameters) maps y_t to 16 state scores, trained with cross-entropy against the true state [2]. It is trained offline (25 minibatches of 300 words, Adam [15], learning rate 10^{-3}) and then takes K Adam steps on each pilot and each accepted data word. The default is $K = 200$; we also report $K = 5$.

VNet-affine: the network is replaced by one affine layer $y_t \mapsto \mathbf{a}y_t + \mathbf{b} \in \mathbb{R}^{16}$ (32 parameters). Up to a state-independent term, the Gaussian log-likelihood $-(y_t - \mu_x)^2/2\sigma^2$ is exactly $(\mu_x/\sigma^2)y_t - \mu_x^2/2\sigma^2$, so this is the smallest free per-state affine metric that contains the true one. (A metric parameterized by the four taps, as in LS-Viterbi, is smaller.)

The learned receivers are trained offline on words drawn from the same COST 2100 tap sequence used for evaluation, which, if anything, favors them.

Adaptation gate. We follow the reference implementation of ViterbiNet: a data word is used for adaptation if the SER of its RS-decoded bits is at most 0.02, and its labels are the RS re-encoded word when that SER is zero and the detector's hard decisions otherwise. Computing that SER requires the transmitted bits, so this gate is an oracle, not a receiver function; it is applied identically to every adaptive receiver. We also evaluate an implementable gate that uses no ground truth: accept a word when the re-encoded decoded word differs from the hard decisions in at most one RS symbol (bounded-distance decoding with two parity symbols), and adapt on the re-encoded word. We re-ran LS-Viterbi, ViterbiNet ($K = 5$ and 200), and VNet-affine with this gate at 7, 10, and 12 dB, from the same offline weights and on the same draws (20–30 repetitions). It accepts fewer words at 7 dB (33% vs. 44% of data words for LS-Viterbi) but labels them more accurately (15% vs. 37% of accepted words carry any label error), and it changes no receiver's SER by more than the 95% confidence interval; LS-Viterbi keeps the lowest SER under either gate.

C. Monte Carlo protocol

Each repetition draws fresh bits and noise for one 125-word frame. The SER is computed on the RS-decoded information bits of the 120 data words (14,400 bits per repetition). Learned receivers use at least 20 repetitions; the perfect-CSI and LS-Viterbi receivers use at least 100 (up to 10^4 and 10^3 , respectively), with the count set from a predicted SER to target about 100 errors. A point with no errors is reported as the one-sided 95% bound $3/N$ for N evaluated bits. The perfect-CSI, LS-Viterbi, ViterbiNet ($K = 5$), and VNet-affine receivers read repetition $r < 200$ from a shared cache of generated (bits, noise) draws, so their first repetitions are paired; we report paired differences over those shared repetitions. The $K = 200$ ViterbiNet sweep ran with the same code on separate GPU machines with independent draws,

so comparisons against it are unpaired. All reported points were generated after we corrected a data-cache defect that had replayed one draw across repetitions; results collected before the fix were discarded, and every reported point shows repetition-to-repetition variance.

III. RESULTS

A. Matched-information baseline

Fig. 1 and Table I compare the receivers from 0 to 14 dB. In point estimate LS-Viterbi has the lowest SER of all no-CSI receivers at every SNR, and it stays within 3–30% of the perfect-CSI SER (e.g., $2.4e-2$ vs. $2.2e-2$ at 7 dB and $2.8e-7$ vs. $2.2e-7$ at 14 dB). ViterbiNet with its default adaptation is 1.3 times worse at 7 dB and 45 times worse at 14 dB. With 5 adaptation steps the gap narrows but remains: 1.07 times at 7 dB, 1.7 times at 12 dB, and 20 times at 14 dB. On the shared draws, LS-Viterbi is significantly better than ViterbiNet with $K = 5$ at 1, 6, 7, and 9–12 dB and than VNet-affine at 0, 2–7, and 11 dB (paired 95% confidence intervals excluding zero), and never significantly worse; at 13–14 dB the paired differences are within noise. We report results up to 14 dB, the last SNR at which every receiver still produced errors within its simulation budget; there the ViterbiNet ($K = 5$), VNet-affine, and LS-Viterbi SERs each rest on 4 error events (95% Poisson interval 0.3–2.6 times the estimate), so the 14 dB ratios are indicative only.

Why LS-Viterbi tracks the bound. With correct labels, the LS estimate from one word of $T = 136$ samples has error covariance $\sigma^2(\mathbf{X}^T \mathbf{X})^{-1}$, and the metric of state x sees noise of variance $\sigma^2(1 + \mathbf{s}(x)^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{s}(x))$. If the shifted-symbol columns of $\mathbf{X}$ are nearly orthogonal, $\mathbf{X}^T \mathbf{X} \approx T \mathbf{I}_L$ and the excess is L/T . We checked this for RS-coded words: averaged over 2,000 random words and the 16 states, $\mathbf{s}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{s} = 0.0301$ against $L/T = 0.0294$ (median condition number of $\mathbf{X}^T \mathbf{X}/T$: 1.39). To first order LS-Viterbi is the perfect-CSI receiver at an SNR lower by

$$\Delta \approx 10 \log_{10}(1 + L/T) = 0.126 \text{ dB}, \quad (2)$$

or 0.129 dB with the exact average for the coded words. Shifting the perfect-CSI curve by Δ predicts SER ratios of 1.03, 1.07, 1.16, and 1.27 at 4, 7, 10, and 12 dB; we measure 1.04, 1.06, 1.13, and 1.30 (Fig. 2). The prediction ignores label errors in accepted words; that it fits this closely suggests that they have little effect and that the remaining gap is the estimation noise of four unknowns. The learned receivers instead drift away from the bound as the SNR grows, reaching 2–4 times at 12–13 dB with $K = 5$ and 13–56 times at 12–14 dB with $K = 200$: a metric fitted by a few stochastic gradient steps carries an error that is invisible at low SNR and dominant at high SNR.

The noisy-CSI baseline explains why ViterbiNet appears strong in the original comparison: with 25% tap error, the Viterbi detector improves only slowly with SNR, from 3.4×10^{-3} at 12 dB to only 1.3×10^{-3} at 14 dB (Table I), because nothing in it corrects the error. ViterbiNet corrects its model from data, and so does LS-Viterbi, with four parameters and one least-squares solve per accepted word.

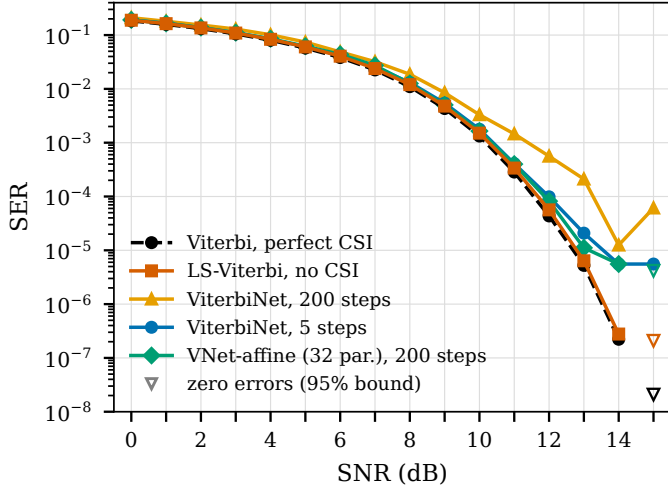


Fig. 1. SER vs. SNR. LS-Viterbi receives the same pilots and accepted decisions as the learned receivers. Results are reported up to 14 dB; hollow triangles at 15 dB mark receivers with no errors (95% upper bound).

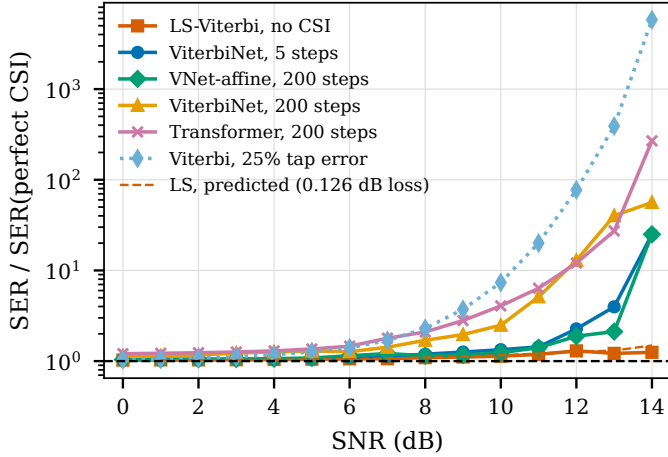


Fig. 2. SER relative to the perfect-CSI Viterbi detector. Dashed red: LS-Viterbi predicted by (2). At 14 dB the $K = 5$, VNet-affine, and LS points rest on 4 error events each.

B. Online adaptation overfits

Fig. 3 sweeps the number of adaptation steps K at 7 dB with 25 offline minibatches. For the 7,002-parameter network the SER is lowest at $K = 5$ (0.0249) and rises steadily to 0.0315 at $K = 200$: repeated steps on a single 136-sample word fit its noise. Without adaptation ($K = 0$) the SER is 0.0340, so adaptation is useful, but a few steps suffice. From 0 to 14 dB, $K = 5$ is better than $K = 200$ at every point (Fig. 1) and costs 40 times less computation per accepted word (Table I).

C. A 32-parameter metric and the latency budget

The affine metric behaves differently: it starts from a generic offline solution and needs many steps to reach the frame's channel, reaching 0.0254 at $K = 100$ –200, statistically tied with the tuned network (Fig. 3). Over the SNR sweep it matches or beats ViterbiNet with $K = 5$ from 8 to 13 dB (Fig. 2). Widening the network does not help: in a sweep of 4 to 64 hidden units and two-layer variants at 7 dB, every larger

TABLE I
SER OF THE RS-DECODED BITS AND ONLINE ADAPTATION COST PER ACCEPTED WORD (ONE CPU THREAD).

Receiver	Param.	7 dB	10 dB	12 dB	14 dB	Adapt.
Viterbi, perfect CSI	–	$2.2\text{e-}2$	$1.3\text{e-}3$	$4.3\text{e-}5$	$2.2\text{e-}7$	–
Viterbi, 25% CSI error	–	$3.8\text{e-}2$	$9.8\text{e-}3$	$3.4\text{e-}3$	$1.3\text{e-}3$	–
LS-Viterbi (no CSI)	4	$2.4\text{e-}2$	$1.5\text{e-}3$	$5.7\text{e-}5$	$2.8\text{e-}7$	LS solve
ViterbiNet, $K=200$	7,002	$3.2\text{e-}2$	$3.3\text{e-}3$	$5.6\text{e-}4$	$1.2\text{e-}5$	209 ms
ViterbiNet, $K=5$	7,002	$2.5\text{e-}2$	$1.8\text{e-}3$	$9.9\text{e-}5$	$5.6\text{e-}6$	5.2 ms
VNet-affine, $K=200$	32	$2.7\text{e-}2$	$1.6\text{e-}3$	$8.2\text{e-}5$	$5.6\text{e-}6$	105 ms

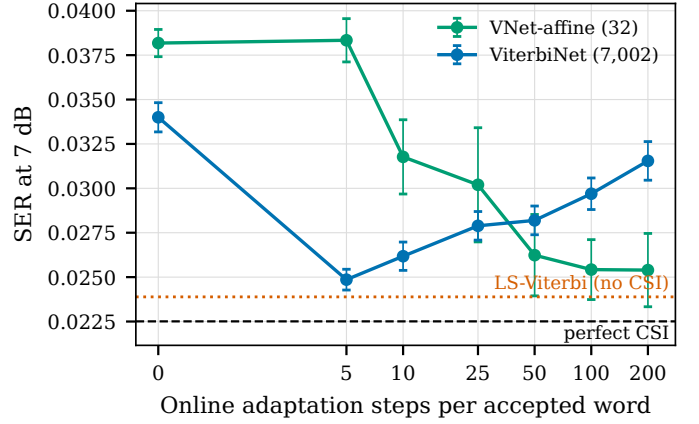


Fig. 3. SER at 7 dB vs. online adaptation steps per accepted word (25 offline minibatches; bars: 95% confidence intervals over 20 repetitions).

model had a higher SER than the affine one at $K = 200$ (point estimates; the smallest networks are within the confidence interval).

The parameter count matters little for latency. On one CPU thread of our implementation, the 136-step, 16-state trellis takes about 11 ms per word ($\approx 85 \mu\text{s}$ per symbol) for every receiver, while the network forward pass takes 0.05 ms (affine) to 0.14 ms (7,002 parameters). The cost that differs is adaptation: one step takes 0.5 ms for the affine metric and 1.0 ms for the network, so the affine metric at $K = 200$ is not cheaper overall than the network at $K = 5$. LS-Viterbi's adaptation is a 4×4 least-squares solve (0.02 ms). At 8 ksymbol/s a word lasts 17 ms; detection plus adaptation of one accepted word takes 11.4 ms for LS-Viterbi and 16.8 ms for ViterbiNet with $K = 5$, but 117 ms for VNet-affine and 220 ms for the default ViterbiNet. The trellis dominates the measured time only because it runs as an interpreted loop; in operation counts it costs $136 \cdot 2^{L+1} \approx 4,400$ add-compare-selects per word, less than one forward pass of the network over the word ($\approx 10^6$ multiply-accumulates), and LS re-estimation costs about 2,200.

IV. DISCUSSION AND CONCLUSION

On this channel the learned branch metric has nothing to discover: the noise is Gaussian and the channel is linear with four taps, so the correct metric is quadratic in y_t with four unknowns. A receiver that estimates those four unknowns from the same data reaches the bound, and a learned metric can at best match it. ViterbiNet's advantage exists only over a

receiver denied that estimate. We therefore recommend that learned detectors be compared against a classical receiver that receives the same pilots and decisions, and that online adaptation budgets be tuned rather than fixed.

The conclusions are limited to the setting studied: BPSK, one COST 2100 tap sequence, a linear Gaussian channel, and our implementation. Learned metrics can still help where the channel is nonlinear or the noise is non-Gaussian, where a four-tap model is wrong; testing those cases against a matched-information baseline is the natural next step.

DATA AVAILABILITY AND AI-USE STATEMENT

Simulation code, raw results, and the scripts that generate every figure and table are available at <https://github.com/Gilzuk/viterbitransformed>. An AI coding assistant (Claude, Anthropic) was used to write simulation and plotting code and to draft the manuscript; the author checked all code, results, and text.

REFERENCES

- [1] N. Shlezinger, J. Whang, Y. C. Eldar, and A. G. Dimakis, "Model-based deep learning," *Proc. IEEE*, vol. 111, no. 5, pp. 465–499, May 2023.
- [2] N. Shlezinger, N. Farsad, Y. C. Eldar, and A. J. Goldsmith, "ViterbiNet: A deep learning based Viterbi algorithm for symbol detection," *IEEE Trans. Wireless Commun.*, vol. 19, no. 5, pp. 3319–3331, May 2020.
- [3] G. D. Forney, Jr., "The Viterbi algorithm," *Proc. IEEE*, vol. 61, no. 3, pp. 268–278, Mar. 1973.
- [4] —, "Maximum-likelihood sequence estimation of digital sequences in the presence of intersymbol interference," *IEEE Trans. Inf. Theory*, vol. 18, no. 3, pp. 363–378, May 1972.
- [5] T. Raviv, S. Park, N. Shlezinger, O. Simeone, Y. C. Eldar, and J. Kang, "Meta-ViterbiNet: Online meta-learned Viterbi equalization for non-stationary channels," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, 2021.
- [6] T. Raviv, S. Park, O. Simeone, Y. C. Eldar, and N. Shlezinger, "Online meta-learning for hybrid model-based deep receivers," *IEEE Trans. Wireless Commun.*, vol. 22, no. 10, pp. 6415–6431, Oct. 2023.
- [7] T. O'Shea and J. Hoydis, "An introduction to deep learning for the physical layer," *IEEE Trans. Cogn. Commun. Netw.*, vol. 3, no. 4, pp. 563–575, Dec. 2017.
- [8] H. Ye, G. Y. Li, and B.-H. Juang, "Power of deep learning for channel estimation and signal detection in OFDM systems," *IEEE Wireless Commun. Lett.*, vol. 7, no. 1, pp. 114–117, Feb. 2018.
- [9] N. Samuel, T. Diskin, and A. Wiesel, "Learning to detect," *IEEE Trans. Signal Process.*, vol. 67, no. 10, pp. 2554–2564, 2019.
- [10] N. Farsad and A. Goldsmith, "Neural network detection of data sequences in communication systems," *IEEE Trans. Signal Process.*, vol. 66, no. 21, pp. 5663–5678, Nov. 2018.
- [11] F. R. Magee and J. G. Proakis, "Adaptive maximum-likelihood sequence estimation for digital signaling in the presence of intersymbol interference," *IEEE Trans. Inf. Theory*, vol. 19, no. 1, pp. 120–124, Jan. 1973.
- [12] R. Raheli, A. Polydoros, and C.-K. Tzou, "Per-survivor processing: A general approach to MLSE in uncertain environments," *IEEE Trans. Commun.*, vol. 43, no. 2/3/4, pp. 354–364, 1995.
- [13] I. S. Reed and G. Solomon, "Polynomial codes over certain finite fields," *J. Soc. Ind. Appl. Math.*, vol. 8, no. 2, pp. 300–304, 1960.
- [14] L. Liu, C. Oestges, J. Poutanen, K. Haneda, P. Vainikainen, F. Quitin *et al.*, "The COST 2100 MIMO channel model," *IEEE Wireless Commun.*, vol. 19, no. 6, pp. 92–99, Dec. 2012.
- [15] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015.